

SHUBHAM PANDEY

Data Scientist

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SUMMARY

Machine Learning Engineer with experience in NLP, Computer Vision, Predictive Analytics, and backend AI deployment. Skilled in building end-to-end ML solutions using Python, SQL, TensorFlow, PyTorch, FastAPI, and AWS across healthcare, finance, and automation domains.

TECHNICAL SKILLS

Languages: Python, SQL | **Databases:** PostgreSQL, Oracle, MySQL | **ML/DL:** Scikit-learn, TensorFlow, PyTorch, CNN, LSTM, Transformers | **Computer Vision:** YOLOv8, OpenCV, OCR | **NLP & LLMs:** Hugging Face, Transformers, Embeddings, NER, Prompt Engineering | **Backend & MLOps:** FastAPI, Django, REST APIs, MLflow, Docker, Git | **Cloud & Tools:** AWS (EC2, S3), Linux, Streamlit, Power BI, Tableau

PROJECTS

Tuberculosis Detection: Python, TensorFlow, CNN, Deep Learning, OpenCV, Streamlit, Explainable AI

- Developed a deep learning-based tuberculosis detection system using chest X-ray images and transfer learning models (VGG16, ResNet50, EfficientNetB0).
- Applied image preprocessing, class imbalance handling, and Grad-CAM explainability to improve model interpretability.

Automatic Number Plate Recognition (ANPR): Python, YOLOv8, EasyOCR, ByteTrack, OpenCV, Computer Vision

- Built an AI-powered Automatic Number Plate Recognition (ANPR) system for real-time vehicle detection, tracking, and OCR.
- Integrated YOLOv8, EasyOCR, and ByteTrack to process images, videos, and live webcam streams.

Object Retrieval using NLP: Python, PyTorch, CNN, LSTM, NLP, Computer Vision, Streamlit

- Developed a CNN-LSTM visual grounding model to localize image regions based on natural language descriptions.
- Implemented phrase encoding, visual feature extraction, and region-ranking mechanisms for object retrieval.

Delivery Time Prediction: Python, Scikit-learn, MLflow, SHAP, Streamlit, Regression

- Developed an end-to-end machine learning pipeline to predict delivery times using historical logistics data.
- Performed EDA, feature engineering, model training, hyperparameter tuning, and experiment tracking with MLflow and implemented SHAP explainability.

PROFESSIONAL EXPERIENCE

Sugmya Finance Pvt. Ltd. | Software Developer

May 2024 – April 2026

- Developed and maintained FastAPI and Django applications serving lending operations across multiple business functions.
- Automated MIS and reporting workflows, reducing manual effort by 40%.
- Designed SQL-based analytics solutions using Oracle and PostgreSQL, supporting business decision-making and operational reporting.

EDUCATION

Masters in Computational Science & Applications — Banaras Hindu University | 2023

B.Tech. Computer Science & Engineering — SHUATS, Allahabad | 2021

CERTIFICATIONS

IIT-M Pravartak (GUVI-HCL) — Master Data Science Program | 2025–Present